

EESHA PEDAKOTA

Visakhapatnam, India | +91-7416413865 | eesha6325@gmail.com | github.com/EESHA-P | [Linkedin](https://www.linkedin.com/in/eesha6325/)

EDUCATION

Vellore Institute of Technology (VIT) — B.Tech, Computer Science and Engineering

August 2026

School of Computer Science and Engineering | CGPA:8.17/10

Coursework : Data Structures and Algorithm, Operating Systems, Computer Networks, Database Management, Programming for Data Science

TECHNICAL SKILLS

Languages: Python, Java, C/C++

AI/ML: TensorFlow, Keras, Transformers/Attention, CNNs (ResNet, MobileNet), Federated Learning (FedAvg), scikit-learn, LLM integration

Backend/Web: FastAPI, REST API design, SQLite, HTML/CSS/JS

Analytics & Visualization : Tableau, PowerBI, Excel, IBM WatsonX, AWS

SoftSkills : Communication, Team Collaboration, Leadership, Problem Solving

INTERNSHIP EXPERIENCE

Qatar Chemical Company Ltd (QChem)-Summer Training Intern

Jun 2025

- Gained hands-on experience with cybersecurity and security monitoring tools, including Nessus, Qualys, Microsoft Defender, Microsoft Sentinel, SIEM, and EDR, supporting vulnerability assessment, threat detection, and security monitoring.
- Leveraged Microsoft Power Platform and Microsoft 365 tools, including SharePoint, Power BI, Power Apps, Power Automate, Power Virtual Agents, and Office 365, for workflow automation, data visualization, collaboration, and reporting.
- Developed foundational experience with enterprise technologies, including SAP ERP, Oracle Database, Enterprise Content Management (ECM), and Visual Studio, supporting data management and enterprise application development.

TECHNICAL PROJECTS

Federated CNN–Transformer Framework for Medical Imaging Classification

Final-Year Capstone Project, VIT (2026) [Github](#)

- Designed Fed-Fusion-T, a federated-learning framework fusing dual CNN backbones (ResNet101V2 + MobileNetV2) with a Transformer encoder to classify brain tumor MRI scans into 4 classes (glioma, meningioma, pituitary, no-tumor) without centralizing patient data.
- Trained and evaluated on 9,388 MRI images (Figshare, SARTAJ, Br35H) and the BRISC dataset, simulating federated clients with FedAvg aggregation over multiple rounds plus fine-tuning epochs in TensorFlow/Keras.
- Achieved 92.75% accuracy / 97.93% AUC-ROC on the Brain Tumor MRI dataset and 95.50% accuracy / 98.90% AUC-ROC on BRISC — within 3–5% of a centralized (non-privacy-preserving) upper bound, and improved F1-score by 2–3% over CNN-only federated baselines via the added attention mechanism.
- Ran a 7-configuration ablation study (centralized vs. federated, single- vs. dual-backbone, with/without Transformer) with confusion-matrix and ROC analysis to isolate each component's contribution.

Smart Parking & Traffic Control Simulation

Team Project (2026) [Github](#)

- Co-built a fully simulated IoT parking system using Node-RED and MQTT for virtual sensor messaging, with a live dashboard for occupancy, dynamic pricing, and rerouting alerts.
- Developed the Python prediction module forecasting 30–60 minute parking availability from historical occupancy data, exposing confidence scores and recommendations.

Code Review Assistant — AI-Powered Static Analysis Tool

Personal Project(2025) [Dashboard](#)

- Built a full-stack tool (FastAPI + HTML/JS dashboard) that uses an LLM (Hugging Face-hosted) to score uploaded code on readability, modularity, and bug risk (1–10) across 12 languages and return categorized, actionable suggestions.
- Implemented a REST API (upload, list, retrieve reviews) with SQLite persistence for review history, and a drag-and-drop web front end for file submission.

Voice Assistant AI — Conversational Voice Agent

Personal Project(2025) [Github](#)

- Built a real-time speech-to-speech assistant in Python on the ElevenLabs Conversational AI API, with configurable agent prompts, live transcript callbacks, and audio I/O handling.

CERTIFICATIONS

- AWS Solutions Architect (as part of coursework) (Nov 2025)
- GEN AI Using IBM Watsonx — IBM Career Education Program / IBM Developer Skills Network (Jun 2025)
- Data Analytics Job Simulation (Data Analysis & Forensic Technology) — Deloitte, via Forage (Aug 2025)